

Strengthening Mechanisms in Steels[☆]

G Krauss, Colorado School of Mines, Golden, CO, USA

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Steels are used for a great number of structural applications, ranging from automotive panels formed from relatively low-strength, high-ductility, low-carbon sheet steels to machine parts and tools heat treated to ultrahigh strengths after forging medium- and high-carbon bar steels. The microstructures that satisfy this wide range of combinations of strength and ductility are produced most often by solidification of liquid steel and subsequent hot or cold work of the solid steel to produce a product of desired shape and size. Steel components are also manufactured directly to shape by casting and powder metallurgy processing. Heat treatments may be applied to make adjustments in properties, either to increase formability for ease of manufacturing complex shapes, or to adjust combinations of final strength and toughness for demanding structural applications.

This article concentrates on the strengthening mechanisms in wrought carbon and low-alloy carbon steels. These steels consist predominantly of iron and small amounts of other elements arrayed in crystal structures of various types. The matrix microstructure at room temperature consists of iron atoms arranged in body-centered cubic (b.c.c.) patterns in crystals of the phase termed ferrite. At high temperatures, the iron atoms are arranged in face-centered cubic (f.c.c.) arrays termed austenite. The atoms of other elements may replace some of the iron atoms, or if the atoms are small enough, may enter interstitial sites between the iron atoms, making the ferrite and austenite solid solution phases. The transformation of austenite to ferrite or other microstructures on cooling accounts for the great variety of strengths attainable in steels (Krauss, 1990a). Other phases, such as carbides, with different crystal structures may be distributed throughout the ferritic or austenitic microstructures. An excellent review of crystal structures is available in the literature (Barrett, 1985).

In all but very small, perfect single crystals of iron, strengthening of steel is controlled by the resistance of the microstructure, when subjected to stress, to the motion of crystal defects known as dislocations. The greater the resistance to dislocation motion, the higher the strength. The various types of dislocation–microstructure interactions are defined as strengthening mechanisms. Thus, the many possible combinations of strength and ductility of steels are produced by building obstacles to dislocation motion into the microstructures of steel, and almost all steels, whether designed for maximum formability or for ultrahigh strength, are dependent on several mechanisms of strengthening. For example, several types of obstacles to dislocation motion are derived from carbon, and consequently small changes in the carbon content of steels may produce profound changes in strengthening behavior.

Figure 1 shows hardness as a function of carbon content and microstructure of steels containing up to about 1.2 wt.% carbon (Krauss, 1990a). The various microstructures, i.e., martensite, tempered martensite, bainite, and pearlite, are described in later sections of this article as required relative to specific strengthening mechanisms. Hardness, measured by the resistance to penetration of an indenter, correlates well with strength and the ability of a microstructure to sustain plastic deformation. The highest-strength microstructures are produced when austenite transforms to martensite, and the lowest-strength microstructures are produced by the transformation of austenite to ferrite and carbide mixtures, or by annealing to produce a microstructure of spheroidized carbides in a matrix of ferrite. Intermediate strengths are produced by thermal processing to produce the transformation of austenite to pearlite or bainite, or by tempering martensitic microstructures. Steels are roughly classified as low-carbon, containing up to about 0.25 wt.% carbon; medium-carbon, containing from 0.25 wt.% to about 0.6 wt.% carbon; and high-carbon steels, containing in excess of 0.6 wt.% carbon.

The following sections describe the various strengthening mechanisms that combine to produce various combinations of strength and ductility as a function of carbon content in steels. Three types of steels are used to illustrate the interaction of

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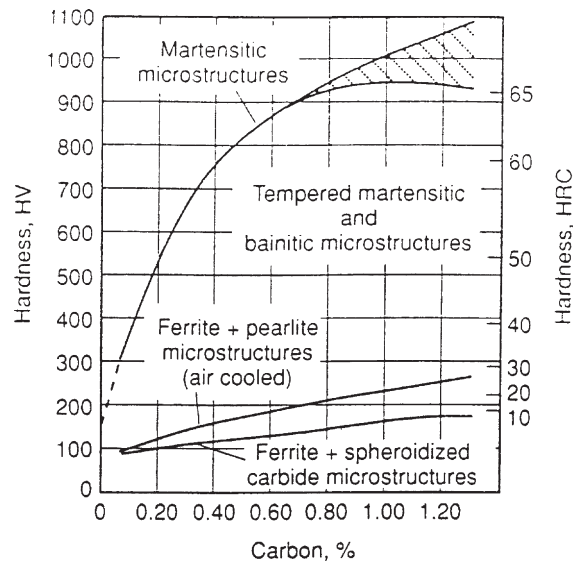


Figure 1 Hardness as a function of carbon content for various microstructures in steels. The cross-hatched area shows the effect of retained austenite in martensitic microstructures (Krauss, 1990a).

dislocations with various microstructural features: (1) low-carbon steels, with largely ferritic microstructures, in which strengthening is controlled by mechanisms associated with strain hardening, aging phenomena, solid solution strengthening, and ferrite crystal sizes; (2) medium-carbon steels, in which carbon takes the form of lamellar carbides alternating with ferrite in a microstructure termed pearlite, and where microalloying, which produces dispersions of alloy nitrides or carbides, provide examples of pearlitic strengthening, and dispersion strengthening; and (3) medium- and high-carbon steels that are heat treated to produce microstructures of tempered martensite in which several strengthening mechanisms combine to produce very high strengths. In the latter steels, some of the parent austenite may be retained, and the deformation-induced transformation of the austenite to martensite constitutes an additional strengthening mechanism.

Prior to a discussion of the strengthening mechanisms that operate in the three types of steels, a brief discussion of the unique behavior of dislocations in the body-centered cubic crystal structure of ferrite is presented.

1 Dislocations and Deformation in Body-Centered Cubic Ferrite

The permanent or plastic deformation of metallic crystal structures is accomplished by slip of planes with close-packed arrays of atoms. The slip of the planes relative to each other is accomplished by dislocation motion which produces localized displacements of the planes, on the order of atomic dimensions, and characterized by a parameter defined as the Burgers vector. As a dislocation glides under sufficiently high shear stresses, the displacement spreads across the slip plane to produce permanent offsets (Cottrell, 1956; Hull and Bacon, 1984). In view of the very small displacements associated with the movement of a single dislocation, huge numbers of dislocations are necessary to produce macroscopically noticeable changes in size and shape of metal parts.

There are many slip systems in b.c.c. ferrite (Hull and Bacon, 1984). Slip planes, in the Miller index notation (Cullity, 1978), are $\{110\}$, $\{112\}$, and $\{123\}$, and the slip direction is in the close-packed $\langle 111 \rangle$ directions. Slip or plastic deformation is accomplished by dislocations with Burgers vectors which join iron atoms in contact along $\langle 111 \rangle$ directions or the body diagonals in the b.c.c. unit cell. Two major types of dislocations develop in crystal structures such as b.c.c. iron. Edge dislocations have Burgers vectors normal to the dislocation line, and are constrained to motion in a given slip plane. Screw dislocations have Burgers vectors parallel to the dislocation line and can cross slip from one slip plane to another. As a result, where cross slip is possible, and in view of the large number of possible slip planes in the b.c.c. structure, microscopically observable slip steps on the surfaces of deformed iron crystals are often irregular, and are described as wavy slip.

The ability of screw dislocations to cross slip around obstacles and participate in dislocation multiplication is responsible for the very high ductilities of steels deformed at or above room temperature. However, with decreasing temperature, cross slip by screw dislocations becomes increasingly difficult, and the stresses required for plastic deformation increase sharply. Figure 2 shows this strong temperature dependence of yielding in essentially pure iron, where the interstitial atoms such as nitrogen and carbon have been effectively removed from solid solution by combination with titanium (Leslie, 1981). Also shown at low temperatures are the elastic limits, σ_E , and anelastic limits, σ_A , the stresses for reversible motion of edge dislocations, and the stresses for long-range motion of edge dislocations, respectively.

The limited ability for screw dislocations to cross slip at low temperatures, based on computer modeling, has been attributed to asymmetries in dislocation core structures (Vitek and Duesbery, 1981). High strain rates also limit the ability for screw-dislocation

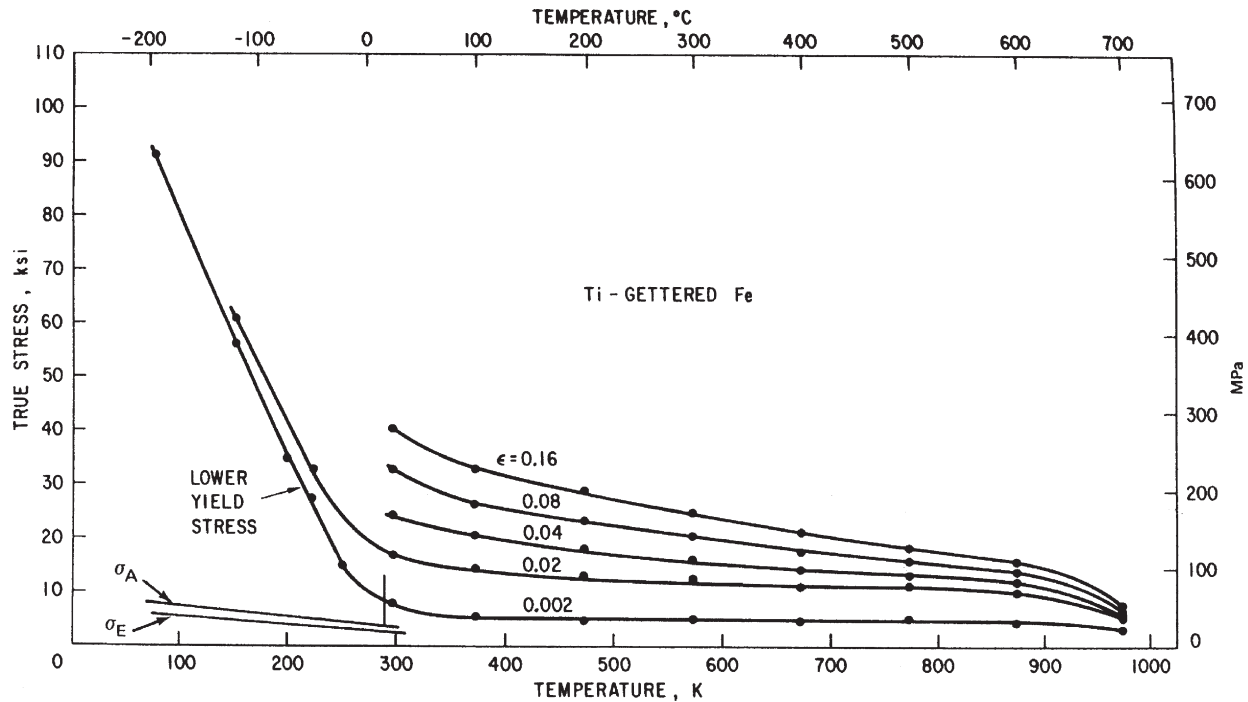


Figure 2 Temperature dependence of yield and flow stresses in titanium-gettered iron with b.c.c. ferrite microstructure (Leslie, 1981).

cross slip, and where dislocation motion is difficult, at low temperature and high strain-rate loading, plastic deformation in b.c.c. structures may be accomplished by deformation twinning. Direct evidence for the restricted cross slip ability of screw dislocations is provided by transmission electron microscopy of thin foils. Long, aligned screw dislocations are observed in b.c.c. ferrite or b.c.c. martensite formed at low temperatures. The interpretation of the latter dislocation substructure is that the edge dislocations have been eliminated by glide and that only screw dislocations, that have not been able to cross slip and multiply to sustain extensive plastic deformation, remain in the ferrite crystals.

The strong temperature dependence and restricted ability for cross slip by screw dislocations results in a severe limitation for the application of steel during low-temperature service. With decreasing temperature, there is a transition from ductile to brittle deformation and fracture behavior. Instead of large amounts of plastic deformation and eventual ductile fracture during loading, at and below room temperature there is limited plastic deformation and fracture occurs by cleavage on $\{100\}$ planes of the b.c.c. structure. The ductile-to-brittle transition may severely limit the use of carbon steels, but low-carbon contents, fine grain sizes, reduced inclusion contents, and large additions of substitutional alloying elements, such as nickel, have been found to significantly lower ductile-to-brittle transition temperatures.

2 Strengthening Mechanisms in Low-Carbon Steels

Low-carbon steels with microstructures consisting primarily of equiaxed crystals or grains of b.c.c. ferrite provide excellent examples of many of the strengthening and deformation mechanisms that operate in steels. These same mechanisms may operate in other types of steels, but other strengthening mechanisms may be superimposed on the mechanisms which dominate the behavior of low-carbon steels, or the steels may be produced to have microstructures substantially different from those of the polygonal ferrite which is the dominant microstructure of low-carbon steels.

2.1 Engineering Stress–Strain Curves

The deformation behavior and mechanical properties, which reflect the strengthening mechanisms operating in a steel, at temperatures where dislocation mobility makes possible measurable plastic strain, are commonly determined by uniaxial tensile testing, where loads are applied parallel to the longitudinal axes of sheet or cylindrical specimens with defined gauge lengths. **Figure 3** shows a schematic engineering stress–strain curve determined by measuring elongation as a function of continuously increasing loads of standardized test specimens. Engineering stress, s , is defined as load divided by the original cross sectional area of a specimen, and elongation or strain is obtained by dividing elongation of the specimen gauge length by its original length. Yield strengths are defined at some level of plastic deformation, typically 0.2%, to establish offset yield strengths, but some dislocation motion and plasticity may develop at strains below the offset strain. Ultimate tensile strength is defined as the

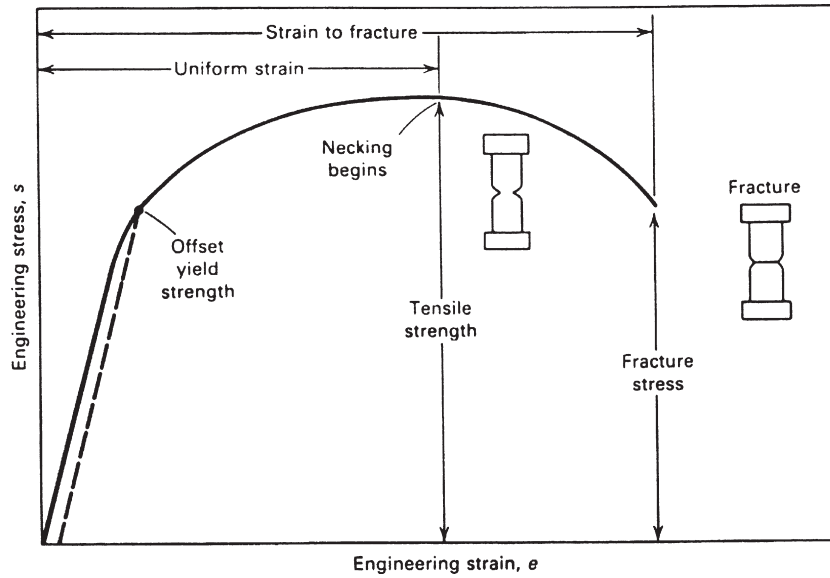


Figure 3 Schematic engineering stress–strain curve established by uniaxial tensile testing (Dieter, 1976).

maximum stress achieved in the test. There are effectively three stages of deformation: yielding, uniform deformation to the ultimate tensile strength, and postuniform, localized deformation to fracture. At the tensile strength, necking instability develops, engineering stress drops, and the remaining deformation occurs within the localized neck, as shown in Figure 3. In low-carbon steels, depending on aging and dislocation density, yielding may be discontinuous or continuous, as discussed in a following section.

2.2 Strain-Hardening Mechanisms

After yielding and the first substantial dislocation movement, the flow strength at a given strain of a steel specimen increases because of work or strain hardening, a strengthening mechanism that results from dynamic dislocation interactions that develop during straining. The shear stress required to move a dislocation line segment and produce plastic flow is inversely proportional to the free length of the dislocation line between pinning points. The pinning points may be sessile junctions due to dislocation intersections or dispersions of second-phase particles. Therefore, as more and more dislocations are generated and interact to sustain plastic flow, the shorter the average effective dislocation-free length or momentary link length, l , and the higher the flow strength at a given level of strain. The momentary flow stress, τ , at any strain during the test, is related to l and the dislocation density ρ by the following equation (Kuhlmann-Wilsdorf, 1985)

$$\tau = \tau_0 + \text{const } Gb/l = \tau_0 + \alpha Gb\rho^{1/2} \quad [1]$$

where G is the shear modulus, b the Burgers vector, α a constant, and τ_0 the friction stress to move a dislocation through the b.c.c. lattice. This equation shows that the effective dislocation link length is inversely related to the square root of the dislocation density, and shows that flow strength will increase as dislocation density increases to sustain plastic deformation.

In low-carbon steels, the dislocations generated during deformation arrange themselves into tangles that become the boundaries for dislocation-free areas or cells within the ferrite grains. The dislocation cell size decreases with increasing strain, reducing the effective dislocation link length within the cell boundaries, consistent with increasing flow stress. Figure 4 shows the decrease in cell size as a function of strain typical of low-carbon steels with ferrite microstructures. The cell size decreases most rapidly in the early stages of deformation, consistent with the highest rates of strain hardening during deformation. Eventually the cell size stabilizes, strain hardening decreases, and the applied stress exceeds the strength of the work-hardened microstructure. At this point, the ultimate tensile strength is attained and mechanical instability or necking develops within a tensile specimen.

Necking instability is defined with the aid of true stress and true strain, which differ from definitions of engineering stress and strain. True stress, σ , is defined as the load divided by the actual cross section of a tensile specimen at a point in a tensile test, and true strain is defined as the natural log of the length of the gauge length at a point in a test divided by the original gauge length. With these definitions, necking instability, which defines the true uniform strain, ϵ_u , occurs when the strain-dependent true strain-hardening rate, $d\sigma/d\epsilon$, equals the strain-dependent true stress, as follows:

$$d\sigma/d\epsilon = \sigma \quad \text{at } \epsilon = \epsilon_u \quad [2]$$

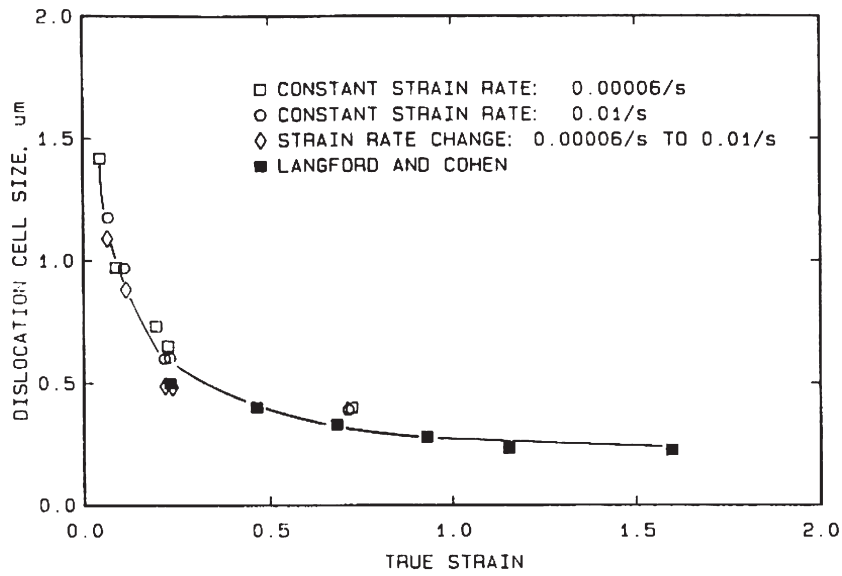


Figure 4 Dislocation cell size as a function of true strain in a low-carbon steel with ferritic microstructure (after Pan in Krauss, 2015).

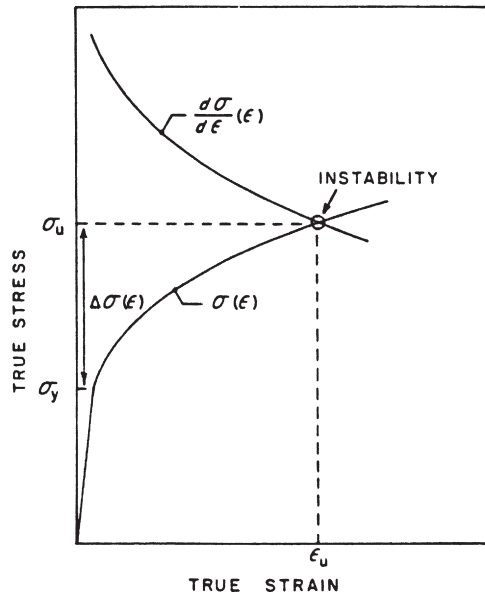


Figure 5 Schematic diagram showing the instability condition (Matlock *et al.*, 1984).

This relationship is shown graphically in [Figure 5](#), where the intersection of the curves of strain hardening and true stress as a function of strain defines necking instability and ϵ_u . Equation [2] offers a useful approach to assess the effect of changes in flow strength levels and strain hardening on the extent of uniform elongation at the ultimate tensile strength. For example, factors that would increase strain hardening, such as finer dislocation substructure, or the strain-induced transformation of retained austenite to a highly dislocated martensite, would cause the intersection of the two curves in [Figure 5](#) to be displaced to higher levels of strain and increased uniform elongation.

2.3 Interstitial Atom Aging Strengthening Mechanisms: Effect on Yielding

[Figure 6](#) shows a schematic diagram of two types of yielding, continuous and discontinuous, that may occur in low-carbon steels. Continuous yielding is a result of the availability of high densities of unpinned dislocations and strain hardening at low plastic strains by mechanisms described above. In low-carbon steels, continuous yielding may be developed by generating high densities of dislocations by mechanical work and limiting the time for aging, or by reducing interstitial carbon contents to very low levels. Another approach consists of heating into the two-phase austenite–ferrite field, forming some volume fraction of austenite in a

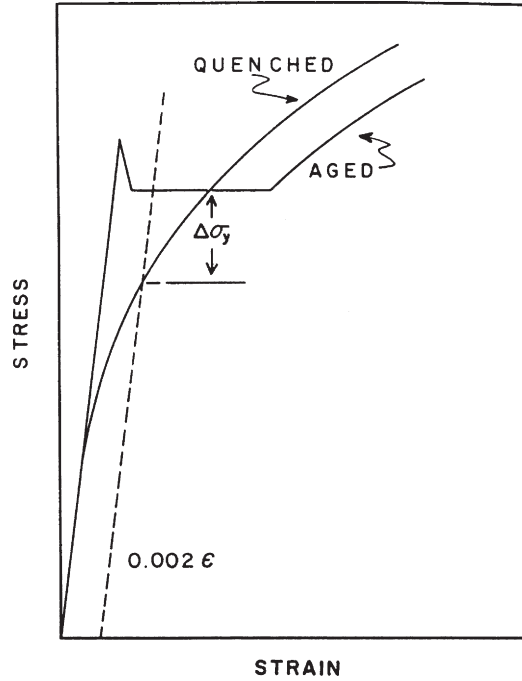


Figure 6 Schematic portions of stress–strain curves to illustrate differences in continuous yielding (quenched curve) and discontinuous yielding (aged curve) (Krauss, 2015).

matrix of ferrite, and then cooling at rates sufficient to transform the austenite to martensite. A high density of unpinned dislocations is formed in the ferrite surrounding the martensite, and these dislocations are available to move and interact immediately upon straining to produce continuous changes in flow stress with increasing strain.

Discontinuous yielding, however, is the rule in low-carbon steels manufactured by hot rolling, cold rolling, and annealing to produce polycrystalline microstructures consisting almost entirely of ferrite and, at most, very small amounts of pearlite, or ferrite and spheroidized carbides, if the steels are cold-rolled and annealed after hot work. These ferritic microstructures provide highly ductile sheet steels for forming automobile panels and appliance applications, but are susceptible to carbon-atom-aging effects that cause discontinuous yielding and surface appearance problems of panels. Carbon atoms produce a tetragonal distortion of the b.c.c. lattice, and can lower their energy by segregating to the atomic discontinuities and elastic strain fields of dislocations. For example, edge dislocations, where the lattice discontinuity may be considered as the line extending along the edge of an incomplete atom plane, have a tensile stress field below that edge, and can accommodate interstitial carbon atoms more readily than octahedral sites in perfect portions of a ferrite crystal (Cottrell, 1956). The diffusion of the carbon atoms to dislocations at room temperature, or on heating to low temperatures, is termed static strain aging.

The pinning of dislocations due to the carbon atom-dislocation interactions is so strong that effectively all of the dislocations are immobilized, and in order to initiate plastic flow, a high density of new dislocations must be generated at a point on the gauge length of a tensile specimen, producing a localized strain or displacement. This strain propagates discontinuously along the gauge length until the entire gauge length is deformed. At that point uniform or continuous deformation by strain hardening proceeds. The strain associated with this process is referred to as the Lüders strain and the displacement that propagates over the gauge length is referred to as a Lüders band. On cold-formed parts subjected to various amounts of deformation, the Lüders strain produces noticeable deformation marks termed stretcher strains, which markedly distort the surface appearance of lightly deformed sheet steels. In order to eliminate the formation of Lüders bands, sheet steels are often temper rolled after annealing to strains exceeding the Lüders strain, so that any plastic deformation associated with subsequent forming will be uniform.

The generation of the high density of dislocations required for discontinuous yielding is initiated at a stress defined as the upper yield strength, and after the Lüders band is nucleated, the stress drops to the lower yield strength, and remains at this level until the band has completely propagated over the gauge length. The drop in stress is explained in part by the stress dependence of dislocation velocity, v , according to the following equation:

$$v = (\tau/\tau_0)^m \quad [3]$$

where τ is the applied shear stress in a slip plane and τ_0 is the shear stress for a constant v of 1 cm s^{-1} . This relationship is illustrated in Figure 7 for edge dislocations at various temperatures in silicon iron (Hull and Bacon, 1984). In a constant strain-rate test,

$$\dot{\epsilon}' = 0.5 b \rho v \quad [4]$$

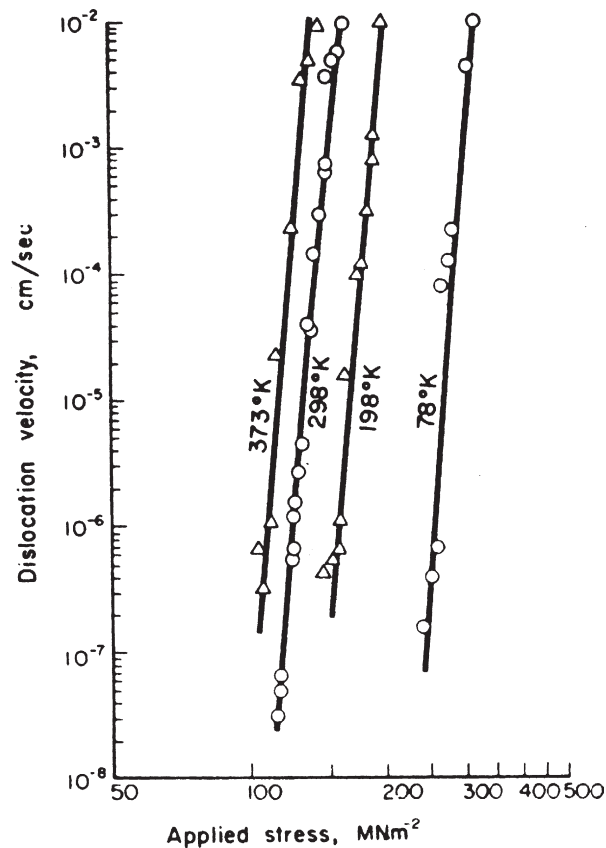


Figure 7 Dislocation velocity as a function of applied stress and temperature in iron containing 3.25% silicon (Hull and Bacon, 1984).

where b is the Burgers vector, ρ the dislocation density, and v the dislocation velocity. Therefore, when the dislocation density discontinuously increases, v must decrease, and according to eqn [3], the stress must drop.

Because of the high mobility of small interstitial nitrogen and carbon atoms in b.c.c. ferrite, even in steels that have been temper rolled to eliminate the Lüders strain, discontinuous yielding may return as carbon atoms diffuse to unpinned dislocations. Figure 8 illustrates the effects of static strain aging in a low-carbon steel. The steel has been strained beyond the Lüders strain and unloaded, and reloaded after aging for various times at 60 °C. Discontinuous yielding returns after very short heating times (Leslie, 1981). Even at room temperature, after several months of storage, discontinuous yielding due to static strain aging may develop in low-carbon steels.

Discontinuous yielding may also be produced in low-carbon steels quenched or rapidly cooled from above the solvus temperature where carbon atoms are fully dissolved in ferrite. Ferrite, which has negligible solubility for carbon at room temperature, is therefore supersaturated with carbon atoms, and on low-temperature heating the carbon atoms segregate and precipitate as carbides, reducing dislocation mobility, and creating the conditions for discontinuous yielding (Leslie, 1981). This process is referred to as quench aging.

Recently, vacuum degassing of carbon steels has made possible the reduction of carbon and nitrogen in steels to very low levels, on the order of 30–50 ppm. Small additions of titanium and/or niobium are made to these steels to tie up the last residual amounts of interstitial atoms, and in these steels, referred to as interstitial-free (IF) steels, or ultralow-carbon steels, discontinuous yielding does not occur (Pradhan, 1990).

The segregation of interstitial atoms to dislocations may also occur during plastic deformation, a phenomenon referred to as dynamic strain aging. The carbon atoms segregate to dislocations as they are formed, and as bursts of new dislocations are generated, serrations in the stress–strain curve develop. Figure 9 shows low-strain portions of the stress–strain curves for a carbon steel tested at various temperatures (Leslie, 1981). The serrated yielding due to dynamic strain aging is most apparent at temperatures between 150 and 250 °C. Also shown is the disappearance of discontinuous yielding during testing at 500 °C. At such high temperatures, the b.c.c. lattice can accommodate the distortion produced by carbon atoms, and the driving force to produce carbon atom atmospheres around dislocations decreases.

2.4 Strengthening by Grain Size Refinement

The now classical work of Hall (1951) and Petch (1953) has resulted in an equation which almost universally, for steels and alloys of many kinds and conditions, describes the dependence of flow and fracture stresses on grain size. The refinement of grain size in

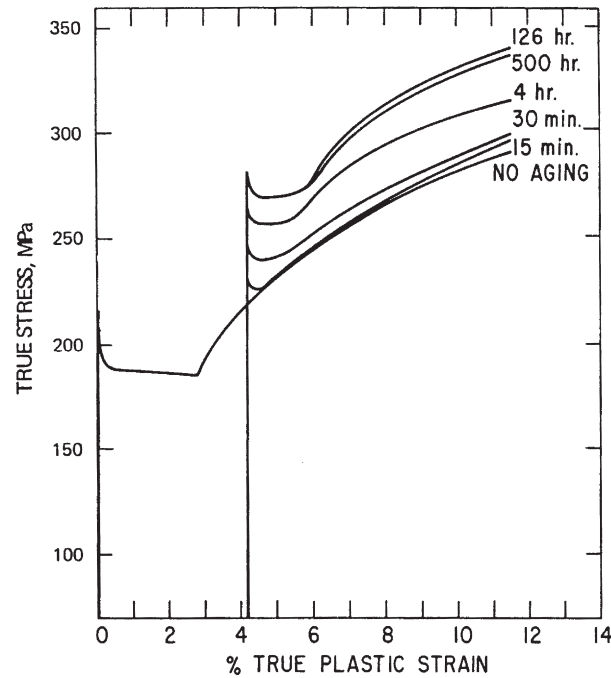


Figure 8 Strain aging effects in a low-carbon steel aged for various times at 60 °C after deformation of slightly over 4% true plastic strain (Leslie, 1981).

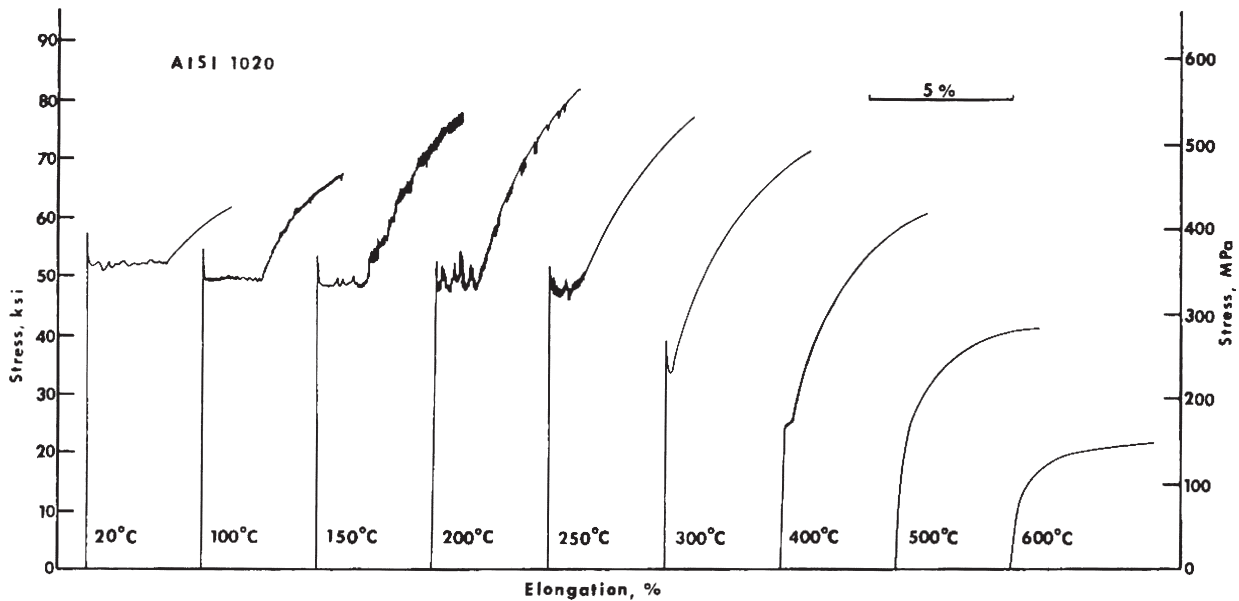


Figure 9 Low-strain portions of stress-strain curves of a low-carbon steel tested at various temperatures as shown (Leslie, 1981).

steels is of considerable practical importance because it provides the one strengthening mechanism which increases both strength and toughness or fracture resistance (Gladman and Pickering, 1983). The Hall-Petch equation for the dependence of a strength parameter σ on ferrite grain size or crystal size, d , is written as follows:

$$\sigma = \sigma_0 + k_y d^{-1/2} \quad [5]$$

where σ_0 and k_y are experimental constants, but for yielding of carbon steels, where grain size is the only variable, they are often taken as the friction stress to move dislocations within the b.c.c. ferrite grains and an unlocking factor for grain boundary dislocations, respectively (Armstrong, 1983).

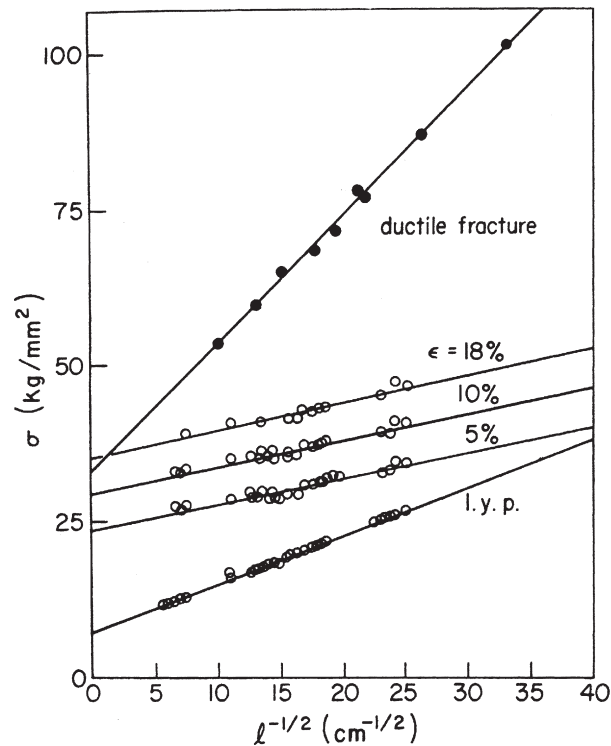


Figure 10 Lower yield stress, flow stresses at various strains, and fracture stress as a function of grain size in low-carbon steels (Armstrong, 1983).

Figure 10 shows a Hall-Petch plot of lower yield stress, flow stresses at various strains, and ductile fracture stress as a function of ferritic grain size in low-carbon steel. All of the strength parameters satisfy the reciprocal square-root dependency on grain size. The slope is steepest for the lower yield strength where because of dislocation locking by carbon atoms, as described above, sufficient stresses must be applied to generate new dislocations. At the higher flow stresses, grain boundaries decrease in importance as the dislocation cell structure develops in the ferrite grains, controlling strengthening. There are several explanations for the strong effect of grain size on strength, including those which state that larger grains permit greater concentrations of dislocations at boundaries and thereby create higher stresses for dislocation movement in adjacent grains, or which assume a unique grain-boundary-dislocation density or structure, such as one in which dislocation sources may be pinned. Nevertheless, the finer the grain size, the greater the fraction of the microstructure affected by grain-boundary phenomena, and the higher the stresses required to activate dislocation motion.

2.5 Substitutional Atom Solid Solution Strengthening

Just as the interactions of interstitial atoms with elastic strain fields of dislocations affect strengthening, atoms that substitute for iron atoms in the b.c.c. ferrite crystal structure may also cause significant strengthening, depending on atomic size and electronic differences. Figure 11 shows the strengthening effect of various elements on the strength of dilute solid solutions of ferrite (Gladman, 1997).

3 Strengthening Mechanisms in Air-Cooled Medium- and High-Carbon Steels

When austenite is cooled from the high temperatures where it is stable, and where the cooling rates are slow enough to permit the diffusion of carbon, the austenite must transform to b.c.c. ferrite, which has very low solubility for carbon. Austenite has a high solubility for carbon, and in the temperature range where it is stable as the only phase, all of the carbon content of the steel is dissolved in the austenite. Therefore, if the carbon content of the steel, and therefore the austenite, exceeds that of ferrite, nominally 0.02 wt.% at the lowest temperature of austenite stability, cementite or iron carbide must form together with the ferrite on cooling. Typically, on slow cooling, ferrite crystals will form first, rejecting carbon into the surrounding parent austenite. Eventually the carbon content of the austenite builds up to the level, nominally 0.8 wt.%, where ferrite crystals grow in parallel to cementite crystals in a two-phase microstructure termed pearlite. The parallel ferrite and cementite crystals are termed lamellae, and the spacing between the crystals is termed the interlamellar spacing. The fine two-phase microstructure of pearlite presents

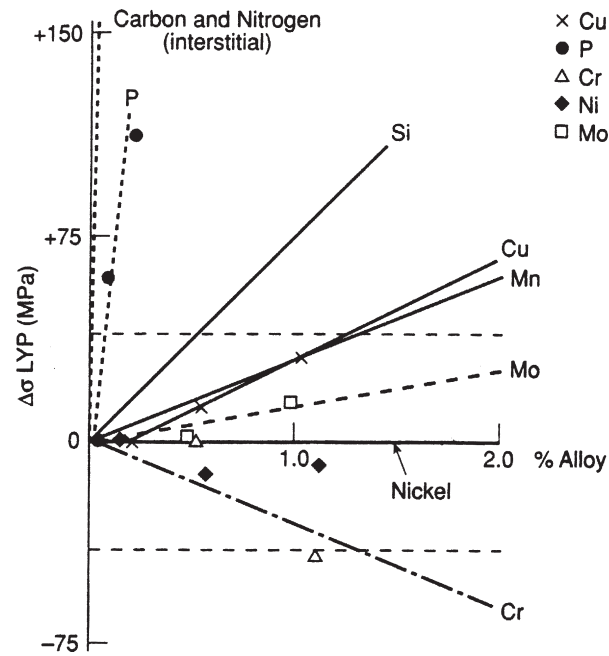


Figure 11 Solid solution strengthening by various alloying elements in b.c.c. ferrite (Gladman, 1997).

additional obstacles to dislocation motion and thereby increases strength. Further strengthening of microstructures with ferrite and pearlite can be produced by microalloying to produce dispersions of fine particles in ferrite.

3.1 Pearlite Strengthening

The interlamellar spacing of pearlite varies depending on the rate of cooling from the austenite phase field, but is typically on the order of 0.1–0.2 μm in size (Krauss, 2015). This refined microstructure contributes significantly to the strength of a steel, and the yield strength σ_y of a steel containing pearlite is given as

$$\sigma = \sigma_0 + k_s s^{-1/2} \quad [6]$$

where σ_0 represents all other strengthening, k_s is the strengthening coefficient, and s is the interlamellar spacing of the pearlite. Equation [6] has the same form as the Hall–Petch equation, eqn [5]. However, the interlamellar spacing of the parallel crystals of ferrite and cementite in pearlite is much finer than that of the grain size in single-phase polycrystalline ferritic microstructures, and therefore the strengthening due to pearlite merits consideration in its own right.

The pearlite content of an air-cooled microstructure ranges from zero in steels containing 0.02% carbon or less to 100% in steels containing 0.8% carbon. Thus, pearlite strengthening increases directly with steel carbon content, and is substantially enhanced by producing very fine interlamellar spacing. In the extreme, fully pearlitic structures are used for demanding applications such as steel rails and ultrahigh-strength tire cord. In the latter application, fine transformed interlamellar spacings are reduced even more by cold drawing.

3.2 Dispersion Strengthening

Small amounts of alloying elements such as vanadium and niobium are added to carbon steels to create dispersions of fine vanadium or niobium carbonitride particles within austenite or ferrite crystals (Gladman, 1997). In low-carbon steels, niobium is added primarily to reduce grain size, because it precipitates as niobium carbonitride particles at temperatures where austenite is stable. As a result, the recrystallization of hot-rolled austenite at low finishing temperatures can be suppressed, and very fine ferrite grains nucleate and grow at the closely spaced boundaries of the unrecrystallized austenite during cooling. However, in medium-carbon steels, vanadium, which precipitates as fine vanadium carbonitride particles at low temperatures with ferrite as it forms from austenite, produces a component of strengthening to ferrite. The vanadium-containing particles precipitate at the interface of ferrite and austenite during cooling, and therefore are characterized as forming by an interphase transformation mechanism. The vanadium carbonitride dispersion strengthening adds to the pearlite strengthening in medium-carbon steels with ferrite–pearlite microstructures.

The strengthening produced by the interaction of dislocations with a dispersion of particles within a matrix phase was first described by Orowan (1948). Figure 12 (Gladman, 1997) shows this process schematically. The arrows represent the forces acting on a dislocation line approaching and then interacting with particles spaced at a separation of L . The dislocation line loops around the

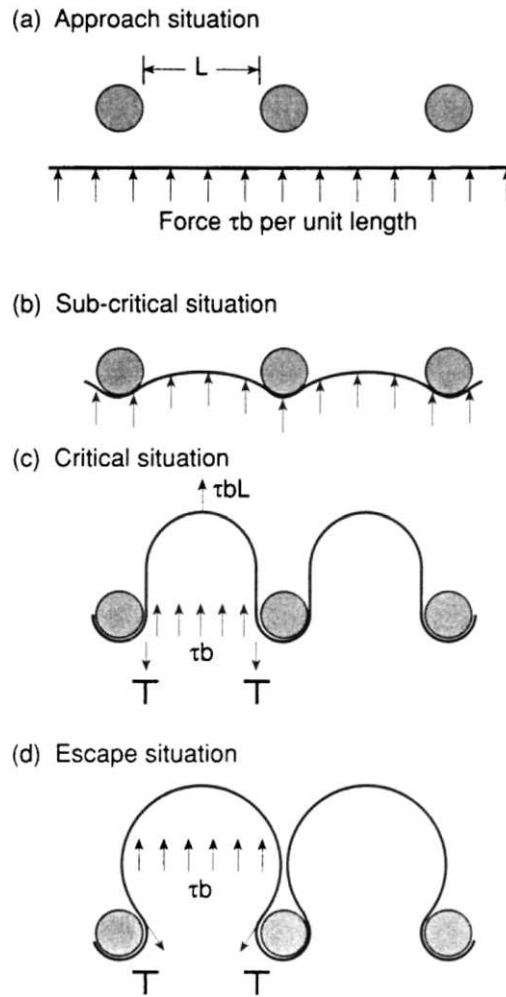


Figure 12 Schematic diagram of stages of Orowan strengthening by interaction of a dislocation with a dispersion of particles with spacing L (Gladman, 1997).

particles and eventually the loops join to form a new dislocation line and residual dislocation loops around the particles. The stress required for this process is inversely proportional to L , and the strengthening increment contributed by particle dispersions is given as

$$\Delta\tau_y = Gb/L \quad [7]$$

where G is the shear modulus and b the Burgers vector. Generally, closely spaced particles on the order of 10 nm in size are required to produce any significant strengthening increment. Equation [7] shows that as L becomes increasingly smaller, significantly higher stresses would be necessary for dislocation motion in the particle dispersions. If the stress becomes too high, the dislocations may actually cut through the particles, rather than looping around the particles as described by the Orowan mechanism.

Figure 13 shows yield strength as a function of carbon content for air-cooled plain-carbon steels, and steels microalloyed with either 0.14 or 0.014 wt.% vanadium and 0.04 wt.% niobium (Sawada *et al.*, 1994). In the plain-carbon steels, the yield strength increases with carbon content are mostly due to increases in the amount of pearlite in the microstructure. The increases in strength due to dispersion strengthening in the microalloyed steels, superimposed on the pearlite strengthening, are also shown.

4 Martensite Strengthening in Medium- and High-Carbon Steels

Martensite is a crystal structure and morphology formed by rapidly cooling or quenching austenite. The rapid cooling prevents carbon atoms from diffusing, and therefore the crystal structure of martensite is body-centered-tetragonal because carbon atoms are trapped between iron atoms and elongate the body-centered unit cell in one direction. In contrast, as noted above, in steels slow-cooled from the austenite, carbon atoms diffuse away from ferrite, and form microstructures consisting of combinations of b.c.c. ferrite and cementite. The latter structures are lower energy and more stable than martensite. Since diffusion is suppressed,

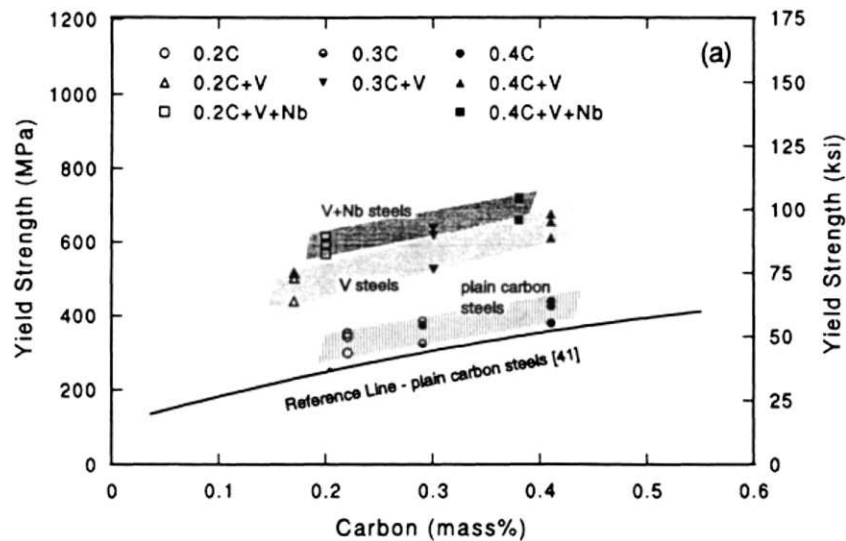


Figure 13 Yield strength as a function of carbon content in plain-carbon steels strengthened by pearlite, and microalloyed steels strengthened by pearlite and dispersion strengthening (Sawada *et al.*, 1994).

martensite crystals are formed by shear mechanisms of transformation. Not only is the composition of the martensite the same as the parent austenite, as it must be in view of the absence of diffusion, but also a very high density of dislocations and/or fine twins are introduced into the martensite crystals to accommodate the shear mechanisms of transformation and volume changes between austenite and martensite.

As quenched martensite tends to have low toughness, and is therefore heated, a process referred to as tempering, to produce more fracture-resistant microstructures (Krauss, 2015). There are many microstructural changes that develop with increasing tempering temperature, but the most important are those associated with the precipitation and growth of carbides to relieve the supersaturation of carbon in the martensite. Thus, carbon diffuses from interstitial sites in the martensite, and the crystal structure reverts to b.c.c. ferrite and dispersions of carbide particles. With increasing tempering temperature, more and more stable and coarser mixtures of ferrite and carbides are formed within as quenched martensitic microstructures.

4.1 Martensitic Strengthening Mechanisms

Figure 1 shows that the highest hardness of any steel microstructure is found in as quenched and in low-temperature-tempered martensite, and that the hardness increases significantly with increasing carbon content. The very high strength of martensite is a result of two interrelated strengthening mechanisms. The one mechanism is the very high rate of strain hardening that develops dynamically from the high-dislocation-density substructure of martensite crystals. The other is dispersion strengthening contributed by the carbides which precipitate in martensite during tempering.

Figure 14 shows the mechanical properties of martensitic low-alloy, carbon steels tempered at 150 °C as a function of carbon content (Krauss, 1995). The separation between the yield strength (YS) and ultimate tensile strength (UTS) show that high rates of strain hardening, increasing with carbon content, develop in these microstructures. The higher rates of strain hardening in the higher-carbon steels are in part related to increasing densities of dislocations and carbide particles precipitated in higher-carbon-content martensite. Strain hardening is also increased by deformation-induced transformation of austenite that has not transformed during quenching. Retained austenite increases in martensitic microstructures with increasing carbon content.

The various measures of ductility, except for uniform elongation, decrease with increasing carbon content because the higher ultimate strengths are close to the stresses that initiate ductile fracture. In lower-carbon steels, where lower ultimate tensile strengths are generated by strain hardening, more postuniform elongation is necessary in order to develop the stresses for ductile fracture. Uniform elongations increase with increasing carbon content as a result of increased strain hardening in the higher-carbon martensites, as shown in Section 2.2 of this article.

Hardness and strengths of martensitic microstructures decrease with heating at increasing tempering temperatures as a result of decreasing dislocation densities and the diffusion-assisted coarsening of the carbide dispersions. As the carbide dispersions coarsen, the interparticle spacings decrease, and lower stresses are required for dislocation motion. At the highest tempering temperatures, around 700 °C, the tempered microstructure has evolved into a dispersion of coarse, spheroidized carbides in a matrix of ferrite, and in this condition, as shown in Figure 1, strength is the lowest of any microstructure in carbon steels.

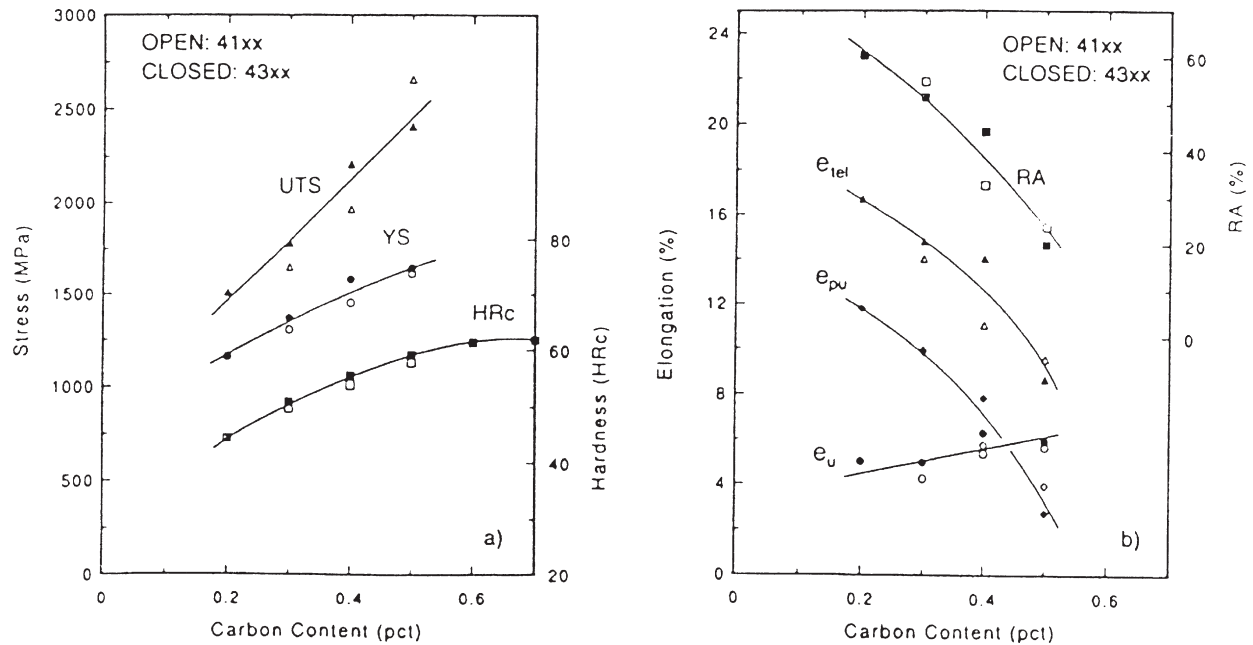


Figure 14 Mechanical properties of low-alloy carbon steels quenched to martensite and tempered at 150 °C: (a) strength properties, (b) ductility properties (Krauss, 1995).

5 Summary

This article has described various strengthening mechanisms associated with the resistance to dislocation motion in carbon steels. Critical to these strengthening mechanisms are the various microstructures in which the mechanisms operate. The reader is referred to the very large literature, only briefly described in this article that describes the design and evolution of the various microstructures by alloying, phase transformations, mechanical deformation, and thermal processing.

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